

Software Design Description 205

ESP32-DevKitV1 – Settings Display Subsystem

Overview

The settings display subsystem gives the user an on-device interface for adjusting system parameters (keyframe send interval, LoRa channel, etc.) without a remote connection. The TFT LCD (480x272, driven by the FPGA) acts as the display. Three tactile buttons on the Tang Nano 9K board (BTN_L, BTN_R, BTN_S) serve as the navigation controls.

Architecture

Hardware Path

BTN_L/R/S (FPGA pins 84/83/85)

- > button_L/R/S wires (top.v)
- > i2c_gpio_expander INPUT_PORT (gpio_in[2:0])
- > DevKit reads via PCA9534 I2C (addr 0x27, reg 0x01)
- > DevKit menu logic decides which setting to change
- > DevKit renders menu to pixel buffer (320x240 grayscale)
- > DevKit streams buffer to FPGA via SPI DMA (fpga_spi_transmit)
- > FPGA BRAM receives pixels (top.v SPI receive path)
- > lcd_controller pat=7 (BRAM mode, default on reset) displays it

Mode Arbitration

Display mode is mutually exclusive with the motion capture pipeline:

- **Capture mode (normal):** PIR active, camera on, FPGA receives image frames, BRAM shows live camera data on LCD.
- **Display mode:** Entered on any button press. Camera and FPGA enables pulled low (via sensors task CAM_EN / FPGA_EN GPIOs). DevKit owns the SPI bus for pixel streaming. Exited by holding BTN_S for 3 seconds, or after a configurable idle timeout.

Button Mapping

Button	Short press	Long press
BTN_L	Previous item / decrement value	–
BTN_R	Next item / increment value	–
BTN_S	Select / confirm	Exit display mode

Settings Items (initial set)

Item	Type	Range	Stored
Keyframe interval	uint32 (hours)	1-24	NVS
LoRa SF	uint8	7-12	NVS
LoRa BW	uint8	7-9	NVS
Stepper home	action	–	–

Display Rendering

The DevKit generates the pixel buffer in SRAM ($320 \times 240 = 76,800$ bytes, DMA aligned). A simple 8x8 bitmap font covers ASCII 0x20-0x7E. The menu renderer writes directly into the buffer; no compositing needed. Each settings change triggers a full-frame SPI transmit (~68 ms at 9 MHz) to refresh the LCD.

No FPGA RTL changes are required. `lcd_controller` starts at `pat=7` (BRAM mode) on reset and stays there unless the physical S2 button (pin 4) cycles it – that button is inaccessible in normal enclosure use.

Implementation Notes

- `fpga_gpio_read_buttons()` (SDD202) already provides debounced reads of `BTN_L/R/S` via PCA9534.
 - NVS namespace: "**settings**", keys match item names above.
 - Settings must be broadcast to the motion pipeline (keyframe interval) and to LoRa init (SF/BW) on exit from display mode.
 - Font table: 95 glyphs x 8 bytes = 760 bytes flash; acceptable.
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Status

PLANNED – not yet implemented.

Dependencies before starting: 1. FPGA pipeline bugs (Bug 1/2/3/4) resolved and validated. 2. Keyframe storage and first-send mechanism in place. 3. NVS read/write utility added to DevKit.